GRADE 7 ACCELERATED · UNIT 4 · ASSESSMENT

Solving Problems with Rational Numbers

Form B · 29 points **WORKED ANSWERS**

Show your work. One point for each lettered part.

Part 1. Operations in Scientific Notation *12 points*

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

1. Add or subtract. Write each answer in scientific notation.

a. $(4.3 \times 10^6) + (2.4 \times 10^6)$ *1 pt*

6.7×10^6

Same power: add the coefficients.

b. $(9.1 \times 10^{-5}) - (3.6 \times 10^{-5})$ *1 pt*

5.5×10^{-5}

c. $(6.4 \times 10^7) + (8.2 \times 10^7)$ *1 pt*

1.46×10^8

14.6×10^7 *is the value; 14.6 is not less than 10, so it is not the answer.*

d. $(5.3 \times 10^5) + (2.6 \times 10^4)$ *1 pt*

5.56×10^5

$2.6 \times 10^4 = 0.26 \times 10^5$.

Scoring: a 1 · b 1 · c 1 · d 1 = 4 points.

2. Multiply or divide. Write each answer in scientific notation.

a. $(4 \times 10^3)(2 \times 10^5)$ *1 pt*

8×10^8

b. $\frac{8.4 \times 10^8}{2.1 \times 10^3}$ *1 pt*

4×10^5

c. $(6 \times 10^{-4})(5 \times 10^6)$ *1 pt*

3×10^3

30×10^2 *is not finished.*

d. $\frac{1.5 \times 10^2}{5 \times 10^6}$ *1 pt*

3×10^{-5}

0.3×10^{-4} *is not finished either.*

Scoring: a 1 · b 1 · c 1 · d 1 = 4 points.

3. Which is the value of $(2.3 \times 10^3) + (4.1 \times 10^5)$? *1 pt*

- ○ A. 6.4×10^5
- ● **B.** 4.123×10^5
- ○ C. 4.33×10^5
- ○ D. 6.4×10^8

B

Scoring: 1 point.

4. Select ALL of the expressions whose value is 4×10^6 . *1 pt*

- ☒ **A.** $(2.5 \times 10^6) + (1.5 \times 10^6)$
- ☐ B. $\frac{1.2 \times 10^{10}}{3 \times 10^4}$
- ☒ **C.** $(8 \times 10^3)(5 \times 10^2)$
- ☒ **D.** $(4.3 \times 10^6) - (3 \times 10^5)$
- ☒ **E.** $\frac{1.2 \times 10^{10}}{3 \times 10^3}$
- ☐ F. $(2.5 \times 10^6) + (1.5 \times 10^5)$

A, C, D and E

C gives 40×10^5 *, which is* 4×10^6 *once it is finished. All four correct and nothing else for the point.*

Scoring: 1 point.

5. Devon writes $(3.4 \times 10^4) + (5.2 \times 10^6) = 8.6 \times 10^{10}$. Name the two mistakes he made, and give the correct sum. *1 pt*

He added the coefficients without first writing both numbers over the same power of ten, and he added the exponents. The correct sum is 5.234×10^6 ***.***

Both mistakes must be named for the point; one of the two is half the item.

$3.4 \times 10^4 = 0.034 \times 10^6$.

Scoring: 1 point.

6. A number is multiplied by 5×10^4 and the product is 3×10^{-2}. What was the number? Write your answer in scientific notation. *1 pt*

6×10^{-7}

$\frac{3 \times 10^{-2}}{5 \times 10^4} = 0.6 \times 10^{-6} = 6 \times 10^{-7}$. *Working a product backwards across a sign change appears on no review and in no question bank; the point is for the finished form, so* 0.6×10^{-6} *alone does not earn it.*

Scoring: 1 point.

Part 2. Real-World Problems in Scientific Notation *4 points*

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

7. Solve. Write each answer in scientific notation. Do not round.

a. A rectangular plate measures 3.5×10^2 mm by 1.25×10^2 mm. Find its area. *1 pt*

4.375×10^4 ***square millimeters***

Multiply the coefficients and add the exponents: 3.5 · 1.25 = 4.375, and $10^2 \cdot 10^2 = 10^4$.

b. A vehicle travels 7.2×10^2 km in 8 hours. Find its average speed. *1 pt*

9×10^1 ***km/h***

$\frac{7.2 \times 10^2}{8} = 0.9 \times 10^2$, *which is not finished: 0.9 is less than 1.*

Scoring: a 1 · b 1 = 2 points.

8. Solve. Write each answer in scientific notation. Do not round.

a. Two lengths are measured: 7.35×10^3 mm and 5.2×10^2 mm. Find their sum. *1 pt*

7.87×10^3 ***mm***

Match the powers first: $5.2 \times 10^2 = 0.52 \times 10^3$ *, and 7.35 + 0.52 = 7.87.*

b. Two masses are measured: 8.4×10^{-2} g and 3.61×10^{-3} g. Find their difference. *1 pt*

8.039×10^{-2} ***g***

Match the powers first: $3.61 \times 10^{-3} = 0.361 \times 10^{-2}$ *, and 8.4 − 0.361 = 8.039.*

Scoring: a 1 · b 1 = 2 points.

Part 3. The Order of Operations with Exponents and Radicals *13 points*

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

9. Evaluate each expression.

a. $20 - 3 \cdot 2^2$ *1 pt*

8

The power first, then the product, then the subtraction.

b. $\frac{4^2 + 9}{5}$ *1 pt*

5

25 over 5; the bar groups the whole numerator.

c. $45 \div 9 \cdot 3$ *1 pt*

15

One level, left to right: 5 · 3. Not 5.

d. $-3^2 - (-3)^2$ *1 pt*

−18

−9 − 9. The parentheses decide what the exponent is attached to.

Scoring: a 1 · b 1 · c 1 · d 1 = 4 points.

10. Evaluate each expression.

a. $\sqrt{8^2 + 17}$ *1 pt*

9

Finish under the bar: 64 + 17 = 81.

b. $\sqrt[3]{-64} + 3^{-2}$ *1 pt*

$-\frac{35}{9}$

$-4 + \frac{1}{9}$.

c. $\left(-\frac{1}{2}\right)^2 + \sqrt{\frac{9}{16}}$ *1 pt*

1

$\frac{1}{4} + \frac{3}{4}$.

Scoring: a 1 · b 1 · c 1 = 3 points.

11. A square patio of side 12 feet is sealed at 2.60 dollars a square foot, with a flat 75 dollars for delivery.

a. Write an expression for the total cost. *1 pt*

$12^2(2.60) + 75$

The rate multiplies the area; the flat fee does not. An expression that charges 75 on every square foot does not earn the point.

b. Find the total cost. *1 pt*

449.40 dollars

144(2.60) = 374.40, then + 75.

Scoring: a 1 · b 1 = 2 points.

12. Which is the value of $3 + 4 \cdot 3^2$? *1 pt*

- ○ A. 27
- ○ B. 63
- ○ C. 225
- ● **D. 39**

D

Scoring: 1 point.

13. Select ALL of the expressions whose value is 8. *1 pt*

- ☒ **A.** $\sqrt{64}$
- ☐ B. $64 \div 4 \cdot 2$
- ☒ **C.** $\left(\frac{1}{8}\right)^{-1}$
- ☐ D. $-\sqrt{64}$
- ☒ **E.** $\frac{5^2+7}{4}$
- ☒ **F.** $\sqrt{6^2+28}$

A, C, E and F

All four correct and nothing else for the point.

Scoring: 1 point.

14. Tomas writes $\sqrt{36+64} = \sqrt{36} + \sqrt{64} = 14$. Explain why he is wrong, and give the correct value. *1 pt*

The radical bar is a grouping symbol, so 36 + 64 is finished before the root is taken: $\sqrt{100} = 10$ ***. The root of a sum is not the sum of the roots.***

The point needs the reason AND the value. '10' alone does not earn it, and neither does 'he added wrong'.

Scoring: 1 point.

15. A square tile has an area of 225 square inches. A second square tile has a side twice as long as the first. What is the area of the second tile? *1 pt*

900 square inches

The side of the first is $\sqrt{225} = 15$ *, so the second has side 30 and area* $30^2 = 900$ *. Doubling a side and asking for the new area appears on no review and in no question bank; a student who doubles the area and writes 450 has not gone through the side.*

Scoring: 1 point.